

Context Free Grammar

→ context free language can describe things that help the context-free grammar (CFG)

Definition:

A CFG is a set of rules / productions used to generate patterns of strings.

Why CFG?

☐ CFG, a more powerful method of describing languages.

☐ This type of grammars can describe certain features that have a recursive structure.

☐ CFG first used in the study of human languages.

☐ Pg-101

☐ CFG occurs in the specification and compilation of programming languages.

→ A grammar for a programming language often appears as a reference of people trying to learn the language syntax.

☐ Most compilers & interpreters contain a component called "parser".

→ that extracts the meaning of a program prior to generating the compiled code or performing the interpreted execution.

Formal Definition of CFG:

A context-free grammar is a 4-tuple (V, Σ, R, S) where

(1) " V " is a finite set called the "variables"

" N " / another name "non-terminals"

(2) " Σ " / " T " is a finite set, disjoint from V , called the "set of terminals"

(3) " R " / " P " is a finite set of rules or productions



; with each rule being a variable and a string of variables and terminals

(4) " S " is the start variable; where $S \in V$

Example:

For the language $\{0^n 1^n \mid n \geq 1\}$;

construct a CFG.

Solution

$$\begin{aligned} \{0^n 1^n \mid n \geq 1\} &= \{0^1 1^1, 0^2 1^2, 0^3 1^3, \dots\} \\ &= \{01, 0011, 000111, \dots\} \end{aligned}$$

CFG

$$S \rightarrow 01 / 0S1$$

ଉତ୍ପାଦନା ନିୟମ

$$S \Rightarrow 01 \mid 0s1$$

S "derive" 01 or $0s1$

Explanation

ଉଦାହରଣ: L.H.S. is a symbol use ଯାହାକୁ S ବୋଲି କୁହାଯାଏ $\rightarrow$ variable

$$V = \{S\}$$

R.H.S. is a variable ଯାହାକୁ ଗୋଟିଏ ଟର୍ମିନାଲ ବୋଲି କୁହାଯାଏ $\rightarrow$ Terminal

$$\Sigma = \{0, 1\}$$

ଉଦାହରଣ: rules ଯାହାକୁ productions

$$\left. \begin{array}{l} S \Rightarrow 01 \\ S \Rightarrow 0s1 \end{array} \right\} \text{ଉଦାହରଣ: rules/productions}$$

start variable ଯାହାକୁ ଗୋଟିଏ ଟର୍ମିନାଲ ବୋଲି କୁହାଯାଏ

$$S = S$$

ଉଦାହରଣ: start variable ଯାହାକୁ ଗୋଟିଏ ଟର୍ମିନାଲ ବୋଲି କୁହାଯାଏ

$$S \in V$$

Exercise

Q) Identify the terminals, non-terminals, start variable, from the following grammar

$$(1) E \Rightarrow E + T \mid T$$

$$T \Rightarrow T * F \mid F$$

$$F \Rightarrow (E) \mid id$$

$$(2) S \Rightarrow (L) \mid a$$

$$L \Rightarrow 2, S \mid S$$

Solution:

(1) Non-terminals, $V = \{E, T, F\}$

Terminals, $T = \{+, *, (,), id\}$

variable 0101 0100 010 0101

start variable, $S = E$

(2) $V = \{S, L\}$

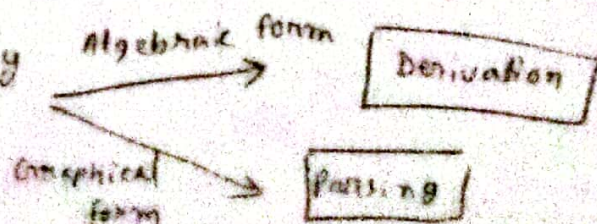
$T = \{(,), ', ', ', a\}$

$S = S$

Note:

For grammar (1) (2) string accept or not

the 2nd way



Derivation format

Ex $S \rightarrow 01 \mid 051$

താഴെ ചില 14 ലെ തിരഞ്ഞെടുത്ത ഗ്രാമറുകൾ "0011" of string generate
 തരൂടെ എന്തെങ്കിലും?

Solution

* താഴെ ചില ഗ്രാമറുകൾ (with derivation) ചില ഉണ്ടാക്കുന്ന string
 തരൂടെ എന്തെങ്കിലും

* Rule ; $S \Rightarrow 051$
 $\rightarrow 0011$

ചില variable നെ replace തരൂടെ ഉണ്ടാക്കുന്ന (with) തരൂടെ string
 തരൂടെ എന്തെങ്കിലും

Ex - "000111" of string എന്തെങ്കിലും?

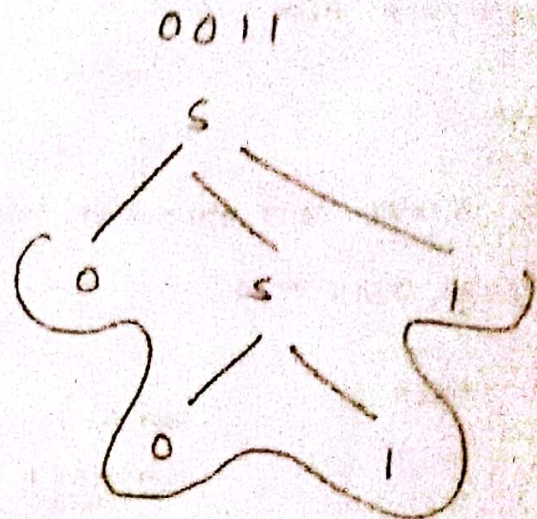
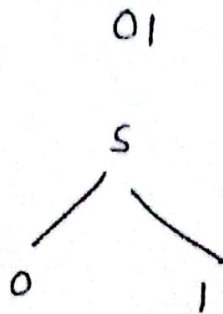
Solution

$S \Rightarrow 051$
 $\Rightarrow 00511$
 $\Rightarrow 000111$

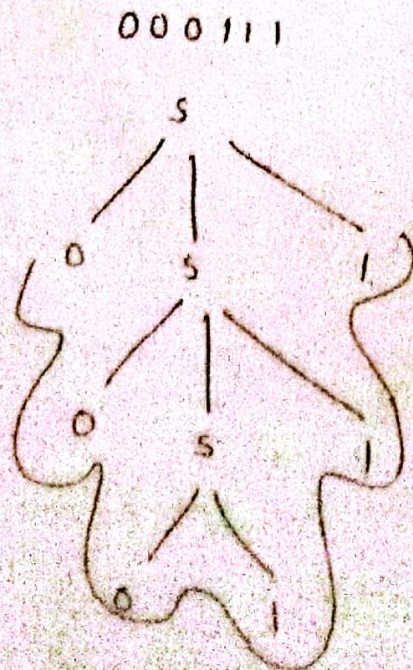
Fahring Festival

$$S \rightarrow 01/0s1$$

Q "0011" string is accepted by the grammar? support me please?



Q "000111" string accepted or not?



CFG design

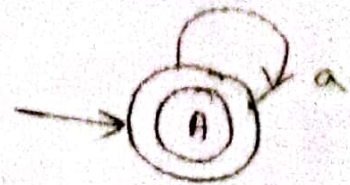
Example - 1

$L = \{a^n \mid n \geq 0\} \rightarrow$ regular expression of this language
 (a^*)

Solution:

$$L = \{a^n \mid n \geq 0\}$$

$$= \{\epsilon, a, aa, \dots\}$$



DFA

CFG

$$A \Rightarrow aA \mid \epsilon$$

in this grammar 'a' and 'A' state a loop and
 when ϵ is used loop ends

checking

" ϵ " accept no?

$$A \Rightarrow \epsilon$$

yes

"aa" accept no?

$$A \Rightarrow aA$$

$$\Rightarrow aaA$$

$$\Rightarrow aa\epsilon$$



so "aa" is generate

with using " ϵ " and no problem

note:

Regular Expression of language

CFG

a^* $\rightarrow$

b^* $\rightarrow$

c^* $\rightarrow$

$A \Rightarrow aA \mid \epsilon$

$A \Rightarrow bA \mid \epsilon$

$A \Rightarrow cA \mid \epsilon$

Example-2

$L = \{a^n \mid n > 1\} \longrightarrow \text{PE } a^+(a^+)$
 $= \{a, aa, aaa, \dots\}$
 one or more

solution:

$A \rightarrow aA \mid a \longrightarrow$ প্রতি "a" মানে accept
 ↓

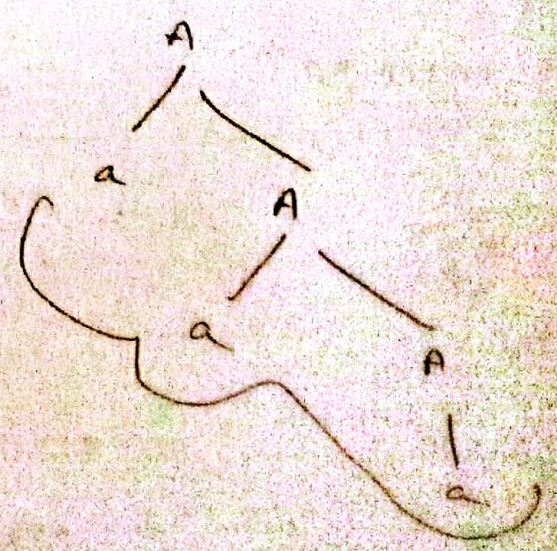
প্রতি মানে "a" আসতে পারে একই state এ
 ফিরে আসবে

checking

"a" accept না?



"aaa" accept না?



Example-3

$L = \{ \text{set of all strings over } a, b \}$ $\longrightarrow$

RE of
 $(a+b)^*$

ଅନ୍ୟ ଲେଖା ଯାଏ

$(a|b)^*$

solution

ଯଦି "a" ଥିବା ଫଳ ମିଳେ

$$A \Rightarrow aA \mid \epsilon$$

ଯଦି "b" ଥିବା ଫଳ ମିଳେ

$$A \rightarrow bA \mid \epsilon$$

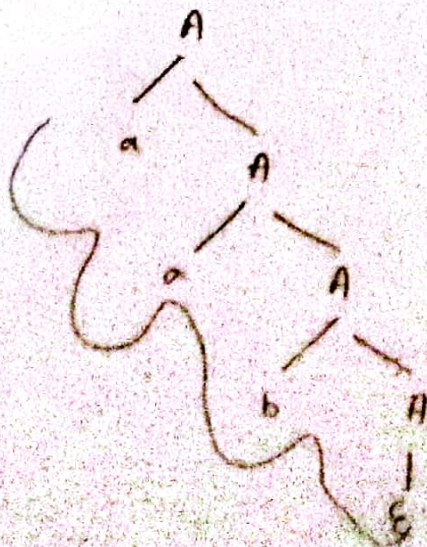
ଉପରୋକ୍ତ ଉଭୟ ଶୃଙ୍ଖଳା $(a+b)^*$ ଯାହା $a^+ b^+$ union

ମାନେ

$$A \rightarrow aA \mid bA \mid \epsilon$$

checking

"aab" string accept କରେ କି?

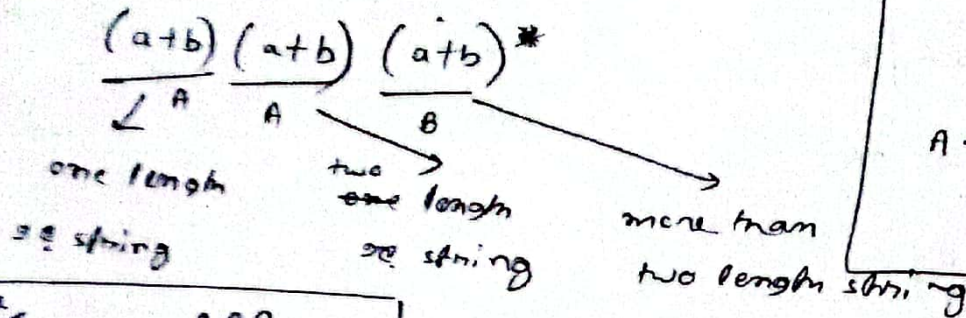


Example-4

CFG for set of all strings which length at least 2

Solution

$$L = \{ aa, ab, ba, bb, aaa, \dots \}$$

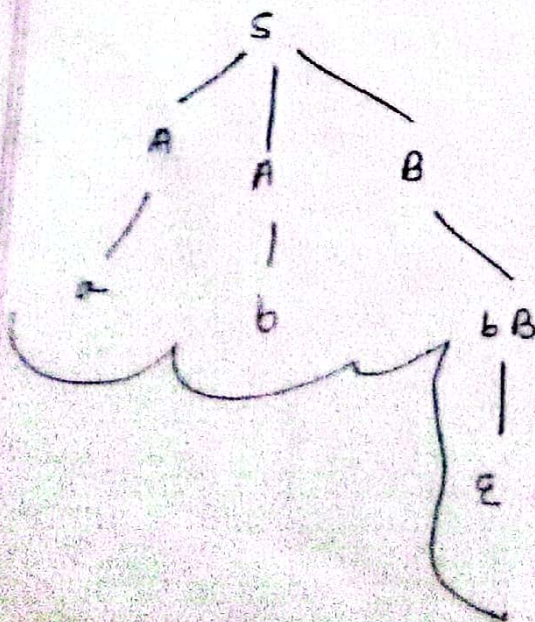


$$\begin{aligned}
 &a^* \\
 &A \rightarrow aA \mid \epsilon \\
 &(a+b)^* \\
 &A \rightarrow aA \mid bA \mid \epsilon
 \end{aligned}$$

CFG

$$\begin{aligned}
 S &\rightarrow AAB \\
 A &\rightarrow a \mid b \\
 B &\rightarrow aB \mid bB \mid \epsilon
 \end{aligned}$$

Q - "abb" string accept or not?



Example 1

Find a regular expression for the set of all strings of at least 3 0's

Solution

0 0 0 0 0 0 0
 $(0+1)^*$ $(0+1)^*$ $(0+1)^*$ $(0+1)^*$

$(0+1)^*$

or

$S \rightarrow 1000000F$

$F \rightarrow 0F \mid 1F \mid \epsilon$

1000

$S \Rightarrow 1000000F$

$\Rightarrow 10000000F$

$\Rightarrow 100000000F$

$\Rightarrow 1000000000F$

$\Rightarrow 10000000000F$

$\Rightarrow 100000000000F$

$\Rightarrow 1000000000000F$

$\Rightarrow 10000000000000F$

$\Rightarrow 100000000000000F$

Example-5

regular set of all strings of at least 3 0's

Solution

$$\underline{E} \underline{0} E \underline{0} E \underline{0} F$$

$$(0+1)^* (0+1)^* (0+1)^* (0+1)^*$$

$$(0+1)^*$$

Proof

$$S \rightarrow E0E0E0F$$

$$F \rightarrow 0F \mid 1E \mid \epsilon$$

1000

$$S \Rightarrow E0E0E0F$$

$$\Rightarrow 1E0E0E0F$$

$$\Rightarrow 1E0E0F0F$$

$$\Rightarrow 1E0E0E0F$$

$$\Rightarrow 100E0F$$

$$\Rightarrow 100E0F$$

$$\Rightarrow 1000F$$

$$\Rightarrow 1000E$$

$$\Rightarrow 1000$$

Example-6

set of all strings which length at most 2

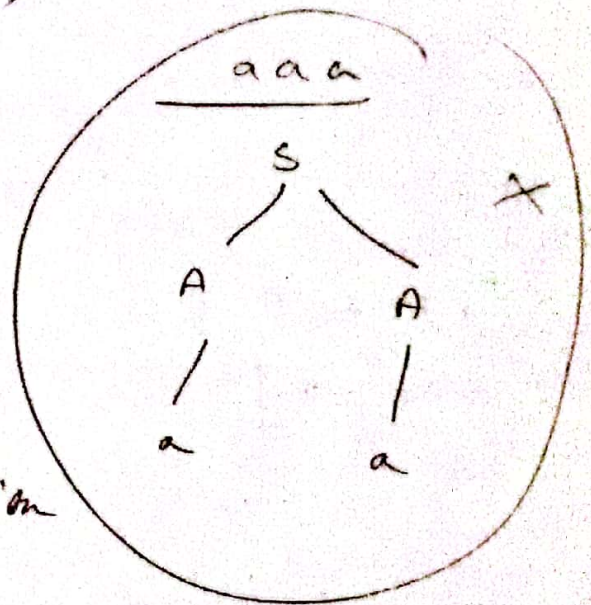
solution

$$L = \{\epsilon, a, b, aa, ab, ba, bb\}$$

$$RE \rightarrow \underbrace{(a+b+\epsilon)}_A \underbrace{(a+b+\epsilon)}_A$$

cfm

$$\begin{aligned} S &\rightarrow AA \\ A &\rightarrow a | b | \epsilon \end{aligned}$$



Example-7

starts with a and ends with

b.

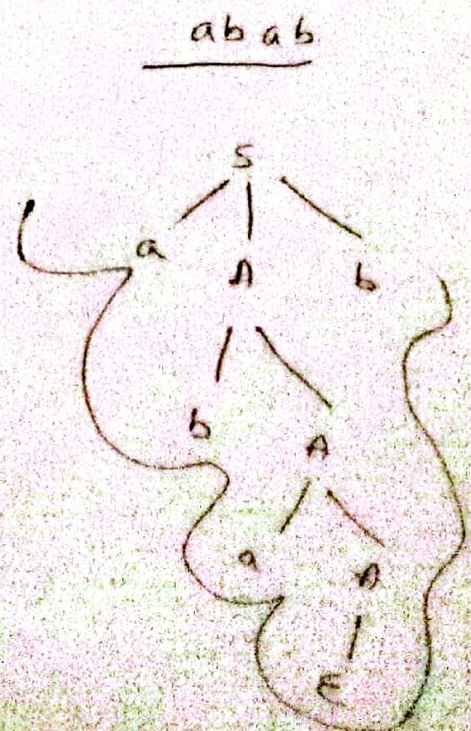
solution

$$a \underbrace{(a+b)^*}_A b$$

cfm

$$S \rightarrow aAb$$

$$A \rightarrow aA | bA | \epsilon$$



Example-8

Starts and ends with different symbol

Solution

$$\frac{a(a+b)^*b}{A} \quad | \quad \frac{b(a+b)^*a}{A}$$

$$a(a+b)^*b + b(a+b)^*a$$

or

$$S \rightarrow aAb \mid bAa$$

$$A \rightarrow aA \mid bA \mid \epsilon$$

Example-9

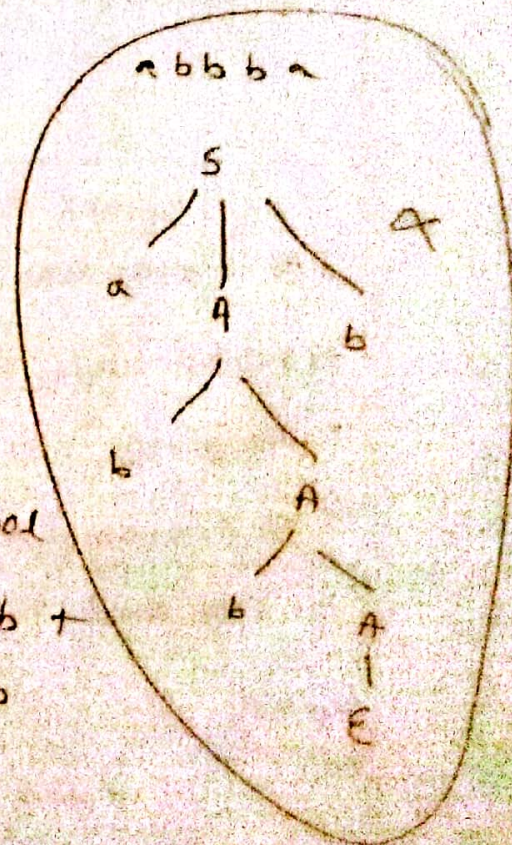
Starts and ends with same symbol

Solution

$$\frac{a(a+b)^*a}{A} + \frac{b(a+b)^*b}{E + a + b}$$

$$S \rightarrow aAa \mid bAb \mid \epsilon \mid a \mid b$$

$$A \rightarrow aA \mid bA \mid \epsilon$$



Example 10

Even length string

$$\left(\frac{(a+b)}{A} \frac{(a+b)}{A} \right)^n$$

$$S \rightarrow BS \mid \epsilon$$

$$B \rightarrow nA$$

$$A \rightarrow a \mid b$$

$$(a+b)^n$$

$$A \rightarrow aA \mid bA \mid \epsilon$$

Ambiguity in CFG

If a grammar generates the same string in several different ways, we say that the string is derived "ambiguously" in that grammar.

If a grammar generates some string ambiguously, we say that the grammar is "ambiguous".

Example

$$S \rightarrow S+S \mid S \times S \mid a$$

$$\begin{array}{l} S \rightarrow S+S \\ S \rightarrow S \times S \\ S \rightarrow a \end{array}$$

pattern/string

$$\text{pattern} \rightarrow a+a \times a$$

Derivation

$$\begin{array}{l} S \rightarrow S+S \\ \rightarrow a+S \\ \rightarrow a+S \times S \\ \rightarrow a+a \times a \end{array}$$

प्रमाण ५ $S \rightarrow S + S$ का अर्थ है कि

$$S \rightarrow S * S$$
$$\rightarrow S + S - S$$
$$\longrightarrow a + a * s$$

→ a f a a

$$\frac{a + a + a}{\downarrow}$$

ambiguously
generated string

1) grammar which is ambiguous generates strings 14213
2) Σ^* "ambiguous grammar".

Remove null productions from
CFG

$$A \longrightarrow E$$

10. ϵ epsilon produce 22 ଓ 23 ମା, null production.

Example - 1

$$S \rightarrow ABC$$
$$A \rightarrow a_n | \varepsilon$$
$$B \rightarrow bB \mid \epsilon$$

0-0

$$A \rightarrow E$$
$$B \rightarrow E$$

Removing $A \rightarrow \epsilon$,

$$S \rightarrow ABC \mid BC$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid \epsilon$$

$$C \rightarrow c$$

Removing $B \rightarrow \epsilon$,

$$S \rightarrow \underline{ABC} \mid \overline{BC} \mid \underline{AC} \mid \underline{\epsilon}$$

ϵ concat

ϵ concat

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

$$C \rightarrow c$$

is grammar for language $a^*b^*c^*$ with
effect $a^*b^*c^*$

$$(2) \quad S \rightarrow ABAC$$

$$A \rightarrow aA | \epsilon$$

$$B \rightarrow bB | \epsilon$$

$$C \rightarrow c$$

$$A \rightarrow \epsilon$$

$$B \rightarrow \epsilon$$

Removing " $A \rightarrow \epsilon$ ",

$$S \rightarrow \underline{ABAC} \mid BAC \mid ABC \mid BC$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid \epsilon$$

$$C \rightarrow c$$

Removing " $B \rightarrow \epsilon$ ",

$$S \rightarrow \underline{ABAC} \mid BAC \mid ABC \mid BC \mid AAC \mid AC \mid c$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

$$C \rightarrow c$$

$$(3) S \rightarrow aS \mid A$$

$$A \rightarrow \epsilon$$

$$A \rightarrow \epsilon$$

$$S \rightarrow \epsilon$$

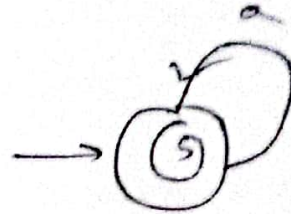
Solution

Removing $A \rightarrow \epsilon$,

$$S \rightarrow aS \mid \epsilon$$

Removing $S \rightarrow \epsilon$,

$$S \rightarrow aS \mid \epsilon \mid a$$



part-2.9;

Remove unit productions
from CFG

Unit Production:

$$A \rightarrow B$$

එනම් variable (වචන) හි දැනට එන
variable produce වේ

Example:

$$A \rightarrow AB$$

$$A \rightarrow a$$

$$B \rightarrow c \mid b$$

$$c \rightarrow d$$

$$d \rightarrow e$$

$$e \rightarrow a$$

solution

for example a grammar $A \rightarrow AB$ (not unit production)

$A \rightarrow a$ (" ")

✓ $B \rightarrow c | b$

[it is not unit production]

✓ $C \rightarrow D$

(C derives D) ["]

✓ $D \rightarrow E$ ["]

$E \rightarrow a$ [not unit production]

steps:

$A \rightarrow AB$

$A \rightarrow a$

$B \rightarrow a | b$

$C \rightarrow a$

$D \rightarrow a$

$E \rightarrow a$

removing unit productions from the L.H.S. gives us the
R.H.S. in the form of

as $C \rightarrow a$
or
 $B \rightarrow a$
or $B \rightarrow b$

as $E \rightarrow a$

or $D \rightarrow E$ so that

or $D \rightarrow a$ is the final

as $D \rightarrow a$

or $C \rightarrow D$ so that

or $C \rightarrow a$ (final)

ଏହି କ୍ଷେତ୍ରରେ $\boxed{\text{unit productions}}$ remove ରିମୁଭ୍ କରାଯାଏ

କିନ୍ତୁ ଏହା check କରାଯାଏ $\boxed{\text{unreachable variables}}$ ଭାବେ ନାହିଁ

Remove unreachable variables:

$A \rightarrow AB$, ଏହି କ୍ଷେତ୍ରରେ $A \rightarrow a$ କ୍ଷେତ୍ରରେ

କ୍ଷେତ୍ରରେ $B \rightarrow a/b$ କ୍ଷେତ୍ରରେ

କିନ୍ତୁ $C \rightarrow a$

$D \rightarrow a$

$E \rightarrow a$

ଏହି କ୍ଷେତ୍ରରେ କ୍ଷେତ୍ର $\boxed{\text{unreachable } A \rightarrow AB}$

କାରଣ ଏହି A କି reach କରା ଯାଏ ନା
କି ଏହି କ୍ଷେତ୍ରରେ ନାହିଁ କି finally

$A \rightarrow AB$

$A \rightarrow a$

$B \rightarrow a/b$

②

$$S \rightarrow aA \mid B$$

$$A \rightarrow ba \mid bb$$

$$B \rightarrow A \mid bba$$

Sol:

Unit productions

$$\left. \begin{array}{l} S \rightarrow B \\ B \rightarrow A \end{array} \right\} \rightarrow S \rightarrow B \rightarrow A$$

Now removing unit productions we get,

$$S \rightarrow aA \mid ba \mid bb \mid bba$$

$$A \rightarrow ba \mid bb$$

$$B \rightarrow ba \mid bb \mid bba$$

checking unreachable variables

Remove unreachable variables

ଯଥା $S \rightarrow \dots$ } ଯଥା ଗ୍ରାମରୁ ଏ B ଯାଏ ତାହା

$$A \rightarrow \dots$$

$$\boxed{B \rightarrow \dots} \text{ ଯାଏ ନାହିଁ}$$

$$S \rightarrow aA \mid ba \mid bb \mid bba$$

$$A \rightarrow ba \mid bb$$

(Ans.)

Part 2.9
(Part 2.10)

Chomsky Normal Form (CNF)

1. Convert Chomsky Normal Form

2. CNF $\rightarrow$ CNF conversion

Step 1: Remove ϵ productions

Step 2 $\rightarrow$ Null production removal

Step 3 $\rightarrow$ Unit

CNF

$\rightarrow$ variable or terminal produce
variable of concatenation produce

1st format 2nd $\boxed{\text{CNF}}$

3rd format
CNF
 $A \rightarrow a$
 $A \rightarrow BC$
 e.g. a either terminal or
 variable of concatenation

Definition:

A context-free grammar is in 'Chomsky Normal Form' if every rule is of the form

$$A \rightarrow BC$$

$$A \rightarrow a$$

where a = any terminal and

A, B, C = any variables - except that A, B, C may not be the start variable

In addition we permit the rule

$S \rightarrow \epsilon$, where S is the start variable

error to CNF conversion

5 steps solution

error format is given below:

① $S \rightarrow ASA \mid aB$

$A \rightarrow B \mid \epsilon$

$B \rightarrow b \mid \epsilon$

solution:

step-1: check that the start variable is R.H.S.

if the start variable does not appear in the R.H.S. then we add a new variable to the R.H.S.

② of example if the start variable is not start variable

if $S \rightarrow ASA$, start variable R.H.S.

$A \rightarrow B \mid \epsilon$ } ϵ appears in the R.H.S. so new variable add to R.H.S.

Add ($S_0 \rightarrow S$)

$S_0 \rightarrow S$

$S \rightarrow ASA \mid aB$

$A \rightarrow B \mid \epsilon$ $B \rightarrow b \mid \epsilon$

Step-2: (Remove Null productions)

අනුව $B \rightarrow \epsilon$ යන null production එය අහ
remove කළ යුතුය

වෙනුවෙන්, $B \rightarrow \epsilon$

නැතහොත්, $A \rightarrow B$
හෝ, $A \rightarrow \epsilon$ } null production

නැතහොත්, $S \rightarrow ASA$

හෝ, $S \rightarrow \epsilon.S.\epsilon$
 $\rightarrow S$

අනුව "A"

නැතහොත් A යනු නැත

"S" නැතහොත් නැත

null වන බැවින්

Removing $B \rightarrow \epsilon$,

$S_0 \rightarrow S$ (නැතහොත් න)

$S \rightarrow ASA / aB / a$

$A \rightarrow B / \epsilon / \epsilon$

අනුව null remove කළ
 aB
 $a\epsilon$ හෙයින් a

B යනු නැතහොත් null නැතහොත්
 B හෙයින් ϵ නැතහොත්

$B \rightarrow b$

[අනුව "ε" remove කළ යුතුය]

Removing $B \rightarrow \epsilon$,

$$S_0 \rightarrow S$$

$$S \rightarrow ASA \mid aB \mid aa$$

$$A \rightarrow B \mid S \mid \epsilon$$

$$B \rightarrow b$$

Removing $A \rightarrow \epsilon$,

$$S_0 \rightarrow S$$

$$S \rightarrow ASA \mid aB \mid a \mid SA \mid AS \mid S$$

↓
உதற்கு AA மிக (மிக)

$$A \rightarrow B \mid S$$

$$B \rightarrow b$$

Step-3: (Removing unit productions)

$$S_0 \rightarrow S, S \rightarrow S, A \rightarrow B, A \rightarrow S$$

அதில் உள்ள unit production

$$S_0 \rightarrow ASA \mid aB \mid a \mid SA \mid AS$$

$$S \rightarrow ASA \mid aB \mid a \mid SA \mid AS \quad [\text{அதில் "S" ஐ B-ஆக, (அதில்) மாற்றுகிறோம்}]$$

$$\cancel{A \rightarrow b} \quad (\text{அதில் } A \rightarrow b \text{ ஐ})$$

$$B \rightarrow b$$

$$A \rightarrow b \mid ASA \mid aB \mid a \mid SA \mid AS$$

Step-4:

ଅନ୍ତିମ CNF ରେ ପରିଣତ କର

ଅନ୍ତିମ ଅବସ୍ଥାରେ check କରାଯାଉଛି

$A \rightarrow a$ (terminal production)
 $A \rightarrow BC$ (two or more variable concatenation)
 (ଉପରୋକ୍ତ ଉପାଦାନ)

$S_0 \rightarrow AX | aB | \underline{a} | SA | AS$

ଅନ୍ତିମ $X = SA$ ରୂପେ

ଅନ୍ତିମ variable (ଅନ୍ତିମ ଅବସ୍ଥାରେ ଉପରୋକ୍ତ ଉପାଦାନ) ରୂପେ

$S \rightarrow AX | aB | \underline{a} | SA | AS$

$A \rightarrow \underline{b} | AX | aB | \underline{a} | SA | AS$

$B \rightarrow b$

$X \rightarrow SA$

Note: Grammar change କରାଯାଇଛି ଯାହା conversion ରୂପେ

Step-5: ଉଦା. $Y \rightarrow a$

$S_0 \rightarrow AX | YB | a | SA | AS$

$S \rightarrow AX | YB | a | SA | AS$

$A \rightarrow b | AX | YB | a | SA | AS$

$B \rightarrow b$

$X \rightarrow SA$

$Y \rightarrow a$